

Aniket Y.

MSc Artificial Intelligence | Uni of Surrey | Agentic AI | RAG | LLM Evaluation | 2+ YoE | Ex-Accenture | AI Engineer
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PROFESSIONAL SUMMARY

Certified Level 7 qualified AI Engineer with 2+ years shipping agentic AI systems in Python, including an FCA-compliance agent achieving 79% accuracy on a curated golden dataset and a live London property-valuation product built from 850,000+ public records and validated by a three-year backtest. Deployed transformer-based entity-resolution models into production at Accenture, improving match accuracy by 25% and predictive model performance by up to 20% across millions of financial records. Awarded Tech-Express Promotion for delivering high-impact analytics solutions.

SKILLS

- **Technical Tools:**
 - **GenAI & LLM Engineering:** LangGraph, LangChain, Model Context Protocol (MCP), RAG Pipelines, Hybrid Retrieval (BM25, Dense Embeddings, Cross-Encoder Reranking), Vector Search (Azure AI Search), LLM Evaluation (Golden Datasets, LLM-as-Judge, RAGAS), Structured Outputs, Prompt Engineering & Injection Defence, Human-in-the-Loop Design, LLM APIs (GPT-4o, Gemini)
 - **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Vision Transformers (SegFormer), Multi-GPU Training, Feature Engineering, Evaluation Metrics, Error Analysis
 - **Programming & APIs:** Python, SQL, Pandas, FastAPI, REST APIs, Pydantic, Git
 - **MLOps & Observability:** Docker, Kubernetes, GitHub Actions, CircleCI (CI/CD), Evaluation-Gated CI, Pytest, Test-Driven Development, LangSmith, OpenTelemetry, Azure Application Insights
 - **Data, Cloud & Apps:** Azure (AI Search, Blob Storage), Elasticsearch APIs, Parquet, Streamlit, Dash, Matplotlib, SAP MM

WORK EXPERIENCE

Independent AI Engineer

Oct 2025 – Aug 2026

- Owned **end-to-end delivery** of **two production AI systems**, from problem framing and **data acquisition** through modelling, evaluation, deployment, and CI, gating every retrieval, prompt, and model change on **automated regression suites**.
- Shipped an **FCA-compliance auditing agent** that automates regulatory review of fintech marketing, achieving **79% verdict accuracy** with a **4% false-flag rate**, and published its FCA Handbook search as an **MCP server**.
- Deployed a live **London property-intelligence product** spanning all **33 boroughs and 850,000+ records**, its valuation signals validated by a **three-year backtest**, serving **LLM-generated briefs** on **Streamlit Cloud**.

Machine Learning Engineer, Accenture

June 2023 – June 2024

- Developed and deployed transformer-based entity resolution models in production using **PyTorch**, **improving matching accuracy by 25%** across large **unstructured banking datasets**, embedding AI/ML capabilities into enterprise financial products supporting **downstream risk modelling** and investment decisioning.
- Built and validated **predictive models across millions of financial records** through feature engineering, cross-validation, and statistical analysis, delivering production AI features that drove enterprise risk at scale.
- Designed modular data ingestion and transformation pipelines (Pandas, Scikit-learn) integrating structured and unstructured enterprise data sources, **reducing ETL processing time by 40%** and producing clean, analysis-ready inputs for downstream AI/ML model development and productionisation.
- Implemented structured testing, validation, and monitoring frameworks across ML workflows, building quality processes for AI output evaluation, **enabling 30% more reliable and faster model deployment** in collaboration with engineering and architecture teams.

Software Engineer, Accenture

April 2022 - June 2023

- Reduced production downtime by **30%** by conducting structured **root cause analysis (RCA)** on SAP MM transactional incidents and supporting resilient production systems.
- Improved reporting accuracy by **10%** by validating supply chain data logic and aligning business rules with financial reporting frameworks, strengthening KPI integrity.

Freelance Full-Stack Web Developer

Jan 2021 – Aug 2022

- Architected and independently shipped **3 full-stack web applications** with end-to-end CI/CD, each serving **20,000+ users**.

PROJECT EXPERIENCE

Sentinel: FCA compliance auditor for fintech marketing

- Built an FCA-compliance auditing agent in **LangGraph** that decomposes fintech marketing copy into discrete claims, retrieves the FCA Handbook rules each claim may breach, and returns cited verdicts, achieving **79% verdict accuracy** with a **4% false-flag rate** on compliant copy.
- Curated a **199-claim golden evaluation dataset** spanning four communication channels before writing pipeline code, verifying every label against the live **FCA Handbook API**, a process that exposed outdated rule numbering in published compliance guides.
- Benchmarked **seven retrieval configurations** across **BM25**, dense embeddings, hybrid fusion, and **cross-encoder reranking**, lifting **recall@5** by **25%** over the BM25 baseline while rejecting the top-scoring configuration as overfit to the **evaluation set**.
- Detected through per-class error analysis that **57%** of genuinely ambiguous claims received overconfident auto-verdicts, gating them behind **human-in-the-loop review**, and published FCA Handbook search as an **MCP server** with **prompt-injection defences**.

Neighbourhood Pulse: Gentrification and valuation gap predictor for London

- Ingested **350,000+ planning applications** and **500,000+ property sales** across all **33 London boroughs** by **reverse engineering** an undocumented **Elasticsearch API**.
- Trained an **XGBoost** house price model (**0.44 R² on held-out data**) that scores every London neighbourhood's **valuation gap** against development signals.
- **Validated the model** with a **backtest**: neighbourhoods flagged undervalued in **2021** **grew 7.3%** over three years while overvalued ones **fell 1.5%**.
- Engineered a **batch LLM pipeline** on the **Gemini API** producing grounded briefs for the **50 most undervalued areas**, enforcing **structured JSON output** through server-side response schemas with **pydantic** re-validation, **prompt guardrails** restricting generation to supplied facts.
- Deployed a **live Streamlit map** and **Dockerised FastAPI** service with **AI-generated** neighbourhood briefs, backed by an automated test suite running in CI.

Multi-Task Vision Transformer for UWF Diabetic Retinopathy(MSc Dissertation – Distinction)

Demonstrates advanced AI modelling, experimentation, and translating complex data insights into actionable outcomes.

- **Engineered a multi-task deep learning pipeline** using PyTorch + SegFormer to jointly perform classification and segmentation, **reducing model parameters by 50% (122M→61M)** while improving deployment efficiency.
- **Optimised predictive performance** by implementing custom training loops with Dice + BCE loss to handle **<0.01% class imbalance**, **achieving 0.97 ROC-AUC and 96% accuracy** across large-scale retinal datasets.
- **Enabled scalable experimentation and model evaluation** via multi-GPU training and systematic hyperparameter tuning, **improving lesion detection to 0.86 F1-score** and surpassing human grader benchmarks.

EDUCATION

MSc Artificial Intelligence, University of Surrey

Sept 2024 – Sept 2025

- Graduated with Distinction; secured 2nd place at Hackathon17 for developing a data-driven prototype solving a real-world business challenge.